YOOSEOK LIM

✉ yooseoklim@umass.edu https://doldolee.github.io

CURRENT POSITION

Seoul National University Hospital Feb 2025 - Present
Researcher, Office of Hospital Information *Seoul, South Korea*
Research Advisor: Hyun-Lim Yang, ByoungJun Jeon

Chungnam National University College of Medicine May 2026 - Present
Researcher, Institute of Biomedical Engineering *Daejeon, South Korea*

EDUCATION

University of Massachusetts Amherst Aug 2026 -
Incoming Ph.D. Student in Biomedical Engineering *MA, USA*
Academic Advisor: Hsiao-Chuan Liu, [Biomedical Ultrasound Lab]

Soongsil University Mar 2022 - Feb 2024
M.S in Industrial and Information Systems Engineering *Seoul, South Korea*
Academic Advisor: Sujee Lee, [AI-Healthcare Lab]
GPA: 3.96/4.0

Soongsil University Mar 2016 - Feb 2022
B.S in Industrial and Information Systems Engineering *Seoul, South Korea*
GPA: 3.69/4.0 (cumulative), 3.68/4.0 (major)
Leave of absence for military service: April 2018 – Nov 2019

PUBLICATIONS AND PREPRINTS

Journal Papers

1. **Y. Lim**, B. Jeon, S. Park, J. Lee, S. Choi, C. Jeong, H. Ryu, H. Lee, H. Yang
   Large Language Model-augmented Offline Reinforcement Learning Framework for Sepsis Management in Critical Care, *NPJ Digital Medicine*, **2026**
2. **Y. Lim**, I. Park, S. Lee
   A Reinforcement Learning Framework for Personalized Anticoagulation Dosing in Critical Care: Integrating Batch-Constrained Policy Optimization and Off-Policy Evaluation, *IEEE Access*, **2025**
3. **Y. Lim**, S. Lee
   OMG-RL: Offline Model-based Guided Reward Learning for Heparin Treatment. *Biomedical Engineering Advances*, **2025**
4. E. Kim, **Y. Lim***
   Mapping Interconnectivity of Digital Twin Healthcare Research Themes through Structural Topic Modeling. *Scientific Reports*, **2025**
5. S. Lee, **Y. Lim**, K. Lim
   Multimodal Sensor Fusion Models for Real-Time Exercise Repetition Counting with IMU Sensors and Respiration Data. *Information Fusion*, **2024**
6. Y. Choe, S. Lee, **Y. Lim**, S. Kim
   Machine Learning-Derived Model for Predicting Poor Post-Treatment Quality of Life in Korean

Cancer Survivors. *Supportive Care in Cancer*, **2024**

In Progress

1. **Y. Lim**, G. Shin, B. Jeon, J. Won, S. Han, H. Yang, M. Lim, J. Leigh
   Trajectory-Level Behaviour Analysis of Age-Related Driving Performance through Inverse Reinforcement Learning, *Under Review*.
2. **Y. Lim**, I. Park, S. Lee
   Trustworthy ICU Care: Enhancing Clinical Safety via Out-of-Distribution Detection-Augmented Offline Reinforcement Learning, *Under Review*.
3. **Y. Lim**, S. Lee
   Generalizable Repetition Counting via Metric-Based Few-Shot Learning on Wearable Motion Signals, *Submitted to IEEE Sensors, Revision*

CONFERENCE PRESENTATIONS

1. **Y. Lim**, I. Park, S. Lee
   Improving Medication Treatment Policies with Batch Deep Reinforcement Learning Algorithms, Conference of the Korean Institute of Industrial Engineers (KIIE), Jeju, Korea, Jun **2023**
2. **Y. Lim,** K. Lim, S. Kim, M. Lee, I, Cho, S. Lee
   Meta-Learning Based Exercise Repetition Counting Framework Using a Head-mounted IMU Sensor, Conference of the Korean Institute of Industrial Engineers (KIIE), Jeju, Korea, Jun **2023**
3. K. Lim, **Y. Lim**, C. Kang, S. Lee
   Establishment of BERT-based Personalized Course Recommendation System, Conference of the Korean Institute of Industrial Engineers (KIIE), Jeju, Korea, Jun **2023**
4. M. Lee, S. Kim, I. Cho, **Y. Lim**, S. Lee
   SSULET: Real-Time Deception Detection Deep Learning Model, Conference of the Korean Institute of Industrial Engineers (KIIE), Jeju, Korea, Jun **2023**
5. **Y. Lim**, S. Lee
   Decision for Optimal Policies on Heparin Treatment using Reinforcement Learning, Conference of the Korea Data Mining Society (KDMS), Busan, Korea, Aug **2022**

RESEARCH PROJECTS

Korean National Police Agency, *South Korea* May 2026 - Present

· Development of an Intelligent (AI)-Based Assessment Algorithm for Evaluating Driving Fitness of High-Risk Drivers and Commercialization of a VR Driving Simulator Platform
  -Role: Project Researcher

Samsung Electronics, *South Korea* Sep 2023 - Feb 2024

· Development of an AI Model and Service Concept based on Canine Activity Data
  -Role: Project Researcher

Soongsil University, *South Korea* Mar 2023 - Dec 2023

· Development of a Course Recommendation System based on Natural Language Data
  -Role: Project Researcher

Samsung Electronics, *South Korea* Jun 2022 - Oct 2022

· Development of a Counting Algorithm based on Human Activity Data
  -Role: Project Researcher

PERSONAL PROJECTS

3D Medical Segmentation

· Implementation of Various 3D Image Segmentation Models [github]

Traffic Sign Detection

· Evaluation of Various Object Detection Model using 2 benchmark datasets [github]

Autonomous Driving Temperature Measurement Robot

· Development of the Autonomous Driving Robot using Computer Vision, Reinforcement Learning, and Edge Computing technologies [github]

SERVICES

Reviews

· Supportive Care in Cancer (2025)
· IEEE Robotics and Automation Letters (IEEE R-AL), Transactions on Industrial Informatics (2024)
· IEEE International Conference on Automation Science and Engineering (IEEE CASE) (2022, 2023)

PATENT

· Driver Fitness Assessment System and Its Control Method, 2026

· Exercise Data Processing Method and Electronic Device Performing the Same.
https://doi.org/10.8080/1020220187795. April. 2024

WORK EXPERIENCE

DataTree
Data Analyst Intern — Jan 2021 - Jun 2021

· Investigating and analyzing a Korean corpus parsing system

Wantreez Music
Software Engineer — Sep 2020 - Dec 2020

· Planning and developing a company CMS website

HONORS AND AWARDS

Scholarships

· BME Departmental Fellowship, University of Massachusetts, Amherst — Spring 2026 - 2027

· Superior Academic Performance Scholarship, Soongsil Univsersity — Spring 2022 - Fall 2023

· College Achievement Scholarship, College of Engineering, Soongil University — Fall 2021

· Superior Academic Performance Scholarship, Soongsil University — Fall 2016

Awards

· Winner, IEEE IUS 2026 QUASOS Challenge — Oct 2026

· Gold Prize, Hyeongnam Science Award — Dec 2021

· Honorable Mention, Hanium Contest — Dec 2021

- Honorable Mention, Hanium Contest — Dec 2020

SKILLS

Programming Language: Python, R, C++, JavaScript, Node JS, Kotlin
Machine Learning: PyTorch, TensorFlow
Others: Raspberry pie, Arduino

RELEVANT COURSEWORK

Mathematics

Probability and Statistics, Linear Algebra, Engineering Mathematics, Machine Learning Mathematics

Machine Learning

Reinforcement Learning, Deep Learning, Data Analysis and its Applications, Data Mining, Data structures and Algorithms, Big data processing, Robot Vision